Devanshi Patel

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EDUCATION

University of Pennsylvania (UPenn), MSE Electrical Engineering — Aug 2025 – May 2027
Digital VLSI Design, Computer Architecture & Organization, Microprocessor Design, Out-of-Order CPU Architecture, Semiconductor Memory Design, System on Chip Architecture

Pandit Deendayal Energy University — July 2021 – May 2025
B.Tech Information and Communication Technology (Electrical Engineering) — GPA: 3.91 / 4.00

EXPERIENCE

Graduate Research Assistant – RISC-V SoC Physical Design | University of Pennsylvania — Feb 2026 – Present
- Run the RTL-to-GDS flow for a Chipyard RISC-V SoC ($\sim$379k cells, $\sim$3$\times$3 mm die, 250 MHz) in a 28nm process via rflowgen, a Rust-based flow generator
- Ran synthesis (Genus), LEC (Conformal), and Innovus floorplan, placement, CTS ($\sim$175k sinks, $\sim$17.9k clock gates), routing, and post-CTS/post-route optimization
- Ran signoff SPEF extraction, Tempus STA, and Voltus static IR; reached post-route setup WNS -1 ps (one reg-to-reg path), hold WNS +36 ps, TNS ≈ 0
- Ran Calibre GDS merge, metal fill, LVS, and DRC, tracing each DRC violation to its design rule and fixing it in the layout or flow setup
- Ran the earlier 40nm implementation (mflowgen) with SDC constraints and multi-corner STA; integrating PrimeTime for signoff timing (in progress)

Electronics Research Intern | eInfochips — May 2024 – Jun 2024
- Designed and simulated a 1-bit Sigma-Delta ADC in MATLAB Simulink applying noise shaping; surveyed mixed-signal ADC architectures and analyzed chip manufacturing data ($>$60nm), gaining exposure to ASIC design flow and semiconductor processes

PROJECTS

Pipelined RISC-V Processor | CIS 5710 | SystemVerilog — Jan 2026 – May 2026
- Designed micro-architecture and RTL implementation of a 5-stage pipelined RISC-V core in SystemVerilog with full RV32I ISA; implemented an AMBA AXI-Lite interface, carry-lookahead adder, integer divider, data forwarding, hazard detection, and branch resolution logic
- Verified with cocotb testbenches, SVA assertions, and RISC-V ISA compliance suites, debugging hazard, forwarding, and branch-resolution bugs at the waveform level; automated regression via TCL/Python scripting on Linux

Optimized Carry Select Adder (OCSA) | Cadence Virtuoso (90nm) (IEEE Publication) — Oct 2024 – Dec 2024
- Designed low-power OCSA datapath using pre-computation and MDCVSL-based multiplexer logic design; reduced propagation delay by 10.52% and power by $\approx$20% for lower PDP; published at IEEE SETCOM 2025

Low-Power Memory: 6T SRAM & NVM (RRAM, STT-MRAM) | Cadence Virtuoso, Verilog-A — Nov 2025 – April 2026
- Designed 6T SRAM array (45nm) with precharge, tri-state drivers, and NAND 4$\times$16 decoder, sizing transistors for read stability and write margin; reduced access delay by 40.2% and power by 23.2% for 54.19% FOM gain
- Integrated Verilog-A NVM compact models (STT-MRAM, RRAM) with custom voltage- and current-mode sense amplifiers; STT-MRAM achieved 31$\times$ lower write energy than RRAM at the bit-cell level and 158$\times$ lower at the 32-bit array level, with sub-10ns read/write latencies for both

Low-Power CNN Hardware Accelerator (FPGA) | SystemVerilog, Xilinx Vivado — Jan 2025 – April 2025
- Designed micro-architecture and RTL of a hardware accelerator for CNN inference on Zybo Z7-20 FPGA; built pipelined fixed-point DSP datapath covering convolution, pooling, and fully connected layers, taking the design from RTL through simulation, synthesis, and on-target verification on hardware
- Applied RTL-level clock gating, 12-bit fixed-point quantization, and memory reuse to meet aggressive power targets; reduced dynamic power by 51.79% and total power by 37.34% while maintaining $>$98% accuracy

RISC-V Out-of-Order SoC with Systolic Array AI Accelerator | SystemVerilog (GitHub) — Jun 2026 – Present
- Designing an RV32I out-of-order core with a 16-entry reorder buffer (ROB), 8-entry reservation stations, register renaming via a register alias table (RAT), and a bimodal branch predictor for out-of-order issue and in-order commit
- Integrating an 8$\times$8 INT8 weight-stationary systolic array over AXI4-Stream; architected an AXI4 crossbar and integrated on-chip SRAM macros for a unified SoC memory interconnect, targeting IHP SG13G2 130nm

Custom Single-Cycle Processor with Novel ISA | Verilog (GitHub) — Dec 2025 – Jan 2026
- Designed custom instruction set architecture and RTL implementation in Verilog with FSM-controlled fetch-decode-execute pipeline, 32-register file, ALU datapath, and status flag handling; verified full ISA execution via Verilog testbenches

SKILLS

Languages & Scripting: TCL, Python, Perl, Bash, SystemVerilog, Verilog, VHDL, Chisel (Chipyard), C/C++, MATLAB, Git, Linux, Verilog-A

Physical Design & Timing: RTL-to-GDS Flow, Logic Synthesis, SDC Constraints, STA, Timing Closure, Place-and-Route, CTS, DRC, PPA Optimization

Circuits & Architecture: CMOS Logic Design, SRAM Design, Low-Power Design, Clock Gating, CPU Microarchitecture, RISC-V ISA, AMBA AXI4/AXI-Lite

EDA Tools: Cadence Genus, Innovus, Xcelium, Virtuoso; Calibre; mflowgen; Chipyard; Xilinx Vivado; cocotb